Milestone 3 — Speaker Companion: SUTVA Statistical Battery

Per-slide guidance, background, and Q&A backup

One-line spine for these two slides. *The economic audit is the SUTVA defense; the permutation test and HLT* confirm *it; the break tests are informative loading-stability diagnostics, not exclusion rules.* If you remember nothing else, say that sentence — it is also exactly what the code now encodes (`config.BREAKPOINT_EXCLUDE` is empty; pool membership follows the audit).

1. How we defend donor selection (SUTVA) [60s]

Talking points

- **Primary defense is the economic audit** — the SCM-canonical approach (Abadie 2021 §3.1: donor exclusion rests on *substantive* grounds, not statistical pretests). Screen every candidate against two contamination channels: the oil-supply path (energy complex, petrocurrencies — categorically out) and the disruption itself (Hormuz routing, Russia/Ukraine trade — donor by donor).
- **Statistical cross-check — the two tests on the slides:**
  - **Permutation mean-shift** (Chow 1960 framing; Lehmann & Romano permutation reference) — the literal SUTVA question (did the donor's own series shift at T_0?), Brent-free, nonparametric. The validation doc designates this as the minimum load-bearing statistical confirmation, and it is the test that produces the lone flag (CNY).
  - **HLT trend break** — Brent-free and I(0)/I(1)-robust, so a break at T_0 *unambiguously* implicates the donor itself. The cleanest of the battery; flags nothing.
- **Bai–Perron and KS are diagnostics, not exclusion rules** — say this explicitly. Bai–Perron regresses on *treated* Brent, so a T_0 break is circular (could just be Brent moving); it measures *loading stability*, not SUTVA. KS over-rejects under macro-regime change. Both are reported but neither auto-excludes a donor.

The other at-T_0 test (run, not on the slides): Wilcoxon event-window

A third at-T_0 test sits in the battery but is kept off the slides:

- **Wilcoxon event-window** (Brown & Warner 1985, event-study tradition): compares the donor's returns in a *short window around* T_0 against a control window, rank-based. Where the permutation test looks for a *persistent mean shift* across the whole pre/post split, the event-window looks for a *localized return reaction* right at the event — they are complementary.
- **Why it is not shown:** on the retained pool it flags *nothing* — it does not even pick up CNY ($p \approx 0.44$ vs the permutation test's $p \approx 0.002$) — and it is lower-powered on the handful of event-window observations. So it adds no information to the result; the permutation test carries the lone flag on its own. Mention it only if asked "did you run more than one at-T_0 test?"
- In the combined BH-FDR flag the pipeline computes, the flag is "permutation *or* event-window"; since event-window fires on nothing here, the combined flag equals the permutation test alone.

Behind the slide

The hierarchy matters: audit is what a reviewer expects and what carries the argument; everything statistical is supplementary. We deliberately do *not* let a test exclude a donor automatically — the test flags a candidate, the researcher decides. This is the "report and interpret" convention of the SCM literature.

If asked

- *Why isn't Bai–Perron a SUTVA test?* It regresses each donor on Brent's returns. Brent is the treated unit, so at T_0 Brent itself jumps; a break in the donor–Brent relationship there cannot be separated from Brent's own move. It tells you whether the donor's *loading* is stable across the fitting window (a parallel-fit concern), not whether the donor was treated.

- *Why drop KS?* It rejects ∼21 of 32 donors on Russia — but that is the macro-regime change (COVID recovery → 2022 tightening), not per-donor treatment. Our own flag rule excludes it for that reason; it is a regime diagnostic, not a SUTVA test.

- *Is HLT definitive?* No — a non-rejection is failure-to-reject, not proof of no interference, and HLT only catches *trend* breaks. It is the cleanest *statistical* triangulation, but the audit remains primary.

2. SUTVA test: no retained donor is treated at T_0 [75s]

The table (verified, 19-donor retained pool)

Flagged at T_0 (of 19 retained)	Hormuz 2026	Russia 2022
HLT trend-break (Brent-free ⇒ SUTVA)	0 / 19	0 / 19
Permutation mean-shift *(on slides)*	0 / 19	1 / 19 (CNY)
Wilcoxon event-window *(run, not shown)*	0 / 19	0 / 19
Bai–Perron (loading stability; appendix)	0 / 19	1 / 19 (VIX)

Talking points

- **HLT flags nothing at either event** — the headline. Since it is Brent-free, this is the strongest clean read: no retained donor's own trend breaks at the event.

- **Two Russia flags, both retained on substantive grounds:**

  - **CNY** (permutation test) — the shift is the concurrent China zero-COVID reopening, not Russia-treatment. The test cannot separate the two; the audit judges it clean.

  - **VIX** (Bai–Perron) — a generic risk-off spike around the invasion, i.e. *intended* common-factor co-movement, which is what SCM exploits, not contamination.

- **Tests inform; the audit decides.** Neither flag triggers an exclusion — that is the policy, and it is why both donors stay in the 19-donor pool.

Behind the slide

VIX was *previously* excluded by an automatic "breakpoint rule." We removed that rule after two checks: (i) the Bai–Perron break does not reproduce when the test is re-run on the analysis window alone (it sits in the trimming dead zone; the contemporaneous early-2022 break is better attributed to HYG), and (ii) a VIX-in/out sensitivity moves the ensemble premium by only −0.2 pp (Russia) and +0.6 pp (Hormuz). Audit-vs-test agreement overall: Russia 18 / 32, Hormuz 31 / 32.

If asked

- *If a test flags VIX, why keep it?* The flag is a loading-stability signal, not SUTVA; it is sample-dependent (does not survive an analysis-window re-run); and it is immaterial to the estimate ($\leq$ 0.6 pp; see `scripts/vix_sensitivity.py`). Excluding it would also force dropping a clean, high-information donor from Hormuz purely to keep a shared pool.

- *Then why show Bai–Perron at all?* Honesty and completeness — it is a legitimate fit-stability diagnostic and it independently re-flags the audit's known cases (e.g. Wheat). We present it as what it is, not as a SUTVA test.

- *Why does the permutation test flag CNY but not the audit?* The test sees "CNY moved at T_0"; it cannot attribute that to Russia versus the concurrent China reopening. The audit, which reasons about causal channels, places CNY's move on the China-policy channel — off the oil-supply / disruption path — so it stays.

- *Couldn't HLT just be under-powered on a short window?* On the full-history spec it is well-powered and clean. A window-restricted HLT does throw a few rejections, but at off-T_0 dates (COVID, 2025 metals moves) — volatility episodes, not focal-event treatment — so the full-history result is the right one to read.